

A Note on “on The Power Dominating Sets of Hypercubes”*

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Abstract

The power domination problem is a variant of the classical domination problem in graphs and is defined as follows. Given an undirected graph $G = (V, E)$, the problem is to find a minimum vertex set $S_p \subseteq V$, called the power dominating set of G , such that all vertices in G are observed by the vertices of S_p . Herein, a vertex observes itself and all its neighbors, and if an observed vertex has all but one of its neighbors observed, then the remaining neighbor becomes observed as well. The minimum cardinality of a power dominating set of a graph is its power domination number. In 2011, Dean et al gave the lower bound and upper bound of power domination number on hypercubes. In this paper, we present the known values of power domination number on Hypercubes and disprove Dean’s conjecture.

Keywords: Hypercubes, Power domination

1 Introduction

Electric power companies need to continually monitor their system’s state as defined by a set of state variables (for example, the voltage magnitude at loads and the machine phase angle at generators [17]). One method of monitoring these variables is to place phase measurement units (abbreviated PMUs) at selected locations in the system. A system is said to be observed if all of the state variables of the system can be inspected by a set of PMUs. Because of the high cost of a PMU, it is desirable to minimize their number while maintaining the ability to observe the entire system.

Let $G = (V, E)$ be a graph representing an electric power system, where a vertex represents an electrical node and an edge represents a transmission line joining two electrical nodes. In [10], a PMU measures the state variable for the vertex at which it is placed and its incident edges and their end vertices (these vertices and edges are said to be observed.) The other observation rules are as follows:

1. Any vertex that is incident to an observed edge is observed.
2. Any edge joining two observed vertices is observed.
3. If a vertex is incident to a total of $k > 1$ edges and if $k - 1$ of these edges are observed, then all k of these edges are observed.

Haynes et al. [10] first considered the graph theoretical representation of the power system monitoring problem as a variation of the well-known graph domination problem (see also [11, 12]). A set $S \subseteq V$ is a *dominating set* in G if every vertex in $V - S$ has at least one neighbor in S . The cardinality of a minimum dominating set of G is the *domination number* $\gamma(G)$. Considering the power system monitoring problem as a variation of the dominating set problem, let a set S_p be a *power dominating set* (abbreviated PDS) if every vertex and every edge in G is observed by S_p . The *power domination number* $\gamma_p(G)$ is the minimum cardinality of a PDS of G . A PDS of G with the minimum cardinality is called a $\gamma_p(G)$ -set. Since any dominating set is a power dominating set, $1 \leq \gamma_p(G) \leq \gamma(G)$ for all graphs G . Haynes et al. [10] also showed that the power domination problem is NP-complete even when restricted to some special classed of graphs such as bipartite graphs or chordal graphs. The power domination problem is widely studied in [1-8, 10, 13, 15-25].

The n -dimensional hypercube Q_n is a graph with 2^n vertices in which each vertex corresponds to an n -tuple $(b_{n-1}, b_{n-2}, \dots, b_0)$ on the set $\{0, 1\}^n$ and two vertices are adjacent by an edge if and only if they differ in exactly one coordinate [14]. For simplicity, we write $b_{n-1}b_{n-2}\dots b_0$ instead of $(b_{n-1}, b_{n-2}, \dots, b_0)$. For example, Figure 1 depicts Q_2 and Q_3 .

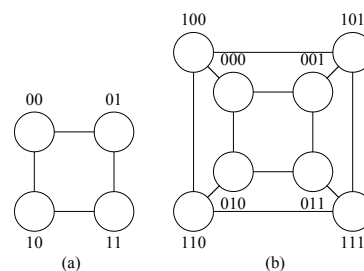


Figure 1: (a) Q_2 ; (b) Q_3 .

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Dean et al [6] gave the lower bound and upper bound of $\gamma_p(Q_n)$. In particular, if $n=2^k$ for some positive integer k , then the lower bound is equal to the upper bound. They conjecture that $\gamma_p(Q_{n+1}) = \gamma(Q_n)$. In this paper, we present some known values of $\gamma_p(Q_n)$ and disprove the conjecture.

2. Preliminaries

All graphs considered here are undirected and simple (i.e., finite, loopless, and without multiple edges). Let $G = (V, E)$ be a graph with vertex set V and edge set E . The open neighborhood $N(v)$ of a vertex v is the set consisting of vertices adjacent to v , i.e., $N(v) = \{u \in V \mid (u, v) \in E\}$, and the closed neighborhood of v is $N[v] = v \cup N(v)$.

In [4, 8, 13, 15], authors found a way to simplify the problem description by using a smaller set of rules equivalent to the four original rules mentioned above.

Observation Rule 1 (abbreviated as OR1):

A vertex in the power domination set observes itself and all its neighbors.

Observation Rule 2 (abbreviated as OR2):

If an observed vertex v of degree $d \geq 2$ is adjacent to $d - 1$ observed vertices, then the remaining unobserved vertex becomes observed as well.

We note that OR1 is just the rule in the definition of dominating set problem. A vertex may dominate vertices at arbitrary distance when certain conditions are fulfilled by OR2 (for example, if a PMU is put in the endvertex of a path, then it can dominate all the other vertices). Also, we shall use the following notations and terms. For $i \in \{0, 1\}$, let Q_n^i denote the subgraph of Q_n induced by the set of vertices with i as the least significant bit. Clearly, each subgraph Q_n^i is isomorphic to Q_{n-1} . The *observed pair* $\langle x, y \rangle$ means that vertex y is the only one unobserved neighbor of vertex x . We use the *observed pair sequence* $(\langle x_1, y_1 \rangle, \langle x_2, y_2 \rangle, \dots)$ to present the order of observed steps. Following this observed pair sequence, all y_i will be observed by OR2.

For example, a path P_6 with 6 vertices is shown in Figure 2. A PDS is located on $\{1\}$ of P_6 . Vertices in $\{0, 2\}$ are observed by OR1. According the order of observed pair sequence $(\langle 2, 3 \rangle, \langle 3, 4 \rangle, \langle 4, 5 \rangle)$, unobserved vertices will be observed by OR2.

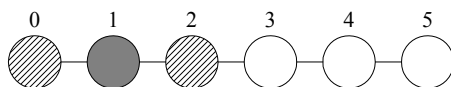


Figure 2: P_6 : The vertex where a PMU is placed is represented by a gray vertex. The vertex is observed by OR1 is filled with slashes.

The power domination problem can be treated as the domination problem with adding a new rule OR2. In [6], the upper bound of power domination number on hypercubes is derived from domination number on hypercubes. We introduce the domination problem on hypercubes as follow.

In [9], Harary and Livingston gave the known values of $\gamma(Q_n)$, as shown in Table 1.

Table 1. The known values of γ for Q_n .

n	1	2	3	4	5	6	2^k-1	2^k
$\gamma(Q_n)$	1	2	2	4	7	12	2^{n-k}	2^{n-k}

Note: integer $k \geq 3$

The following lemma provides an induction to derive the upper bounds of $\gamma(Q_n)$.

Lemma 1 $\gamma(Q_n) \leq 2\gamma(Q_{n-1})$ for $n \geq 2$.

Proof. Suppose that Q_{n-1} has a domination set S with $\gamma(Q_{n-1})$. Since each subgraph Q_n^0 and Q_n^1 is isomorphic to Q_{n-1} , any one of Q_n^0 and Q_n^1 has a domination set S with $\gamma(Q_{n-1})$. Then, $\gamma(Q_n) \leq 2\gamma(Q_{n-1})$ for $n \geq 2$. $\square$

Corollary 2 $\gamma(Q_n) \leq 2^{n - \lfloor \log n \rfloor}$, for $n \geq 9$.

Proof. In Table 1, that $\gamma(Q_n) = 2^{n - \lfloor \log n \rfloor}$ is the base, $n \in \{2^k-1, 2^k\}$, and integer $k \geq 3$. Applying Lemma 1 recursively, we have that $\gamma(Q_n) \leq 2^{n - \lfloor \log n \rfloor}$. $\square$

For example, $\gamma(Q_8) = 32$ in Table 1. By Lemma 1, $\gamma(Q_9) \leq 2\gamma(Q_8) = 2 \times 32 = 64$. Similarly, $\gamma(Q_{10}) \leq 2\gamma(Q_9) \leq 4\gamma(Q_8) = 128$.

Dean et al [6] gave the lower bound and upper bound of $\gamma_p(Q_n)$ and a conjecture as follow.

Theorem 3 $2^{n-1}/n \leq \gamma_p(Q_n) \leq \gamma(Q_{n-1}) \leq 2^{n - \lfloor \log n \rfloor - 1}$. [6]

Conjecture 4 $\gamma_p(Q_{n+1}) = \gamma(Q_n)$. [6]

3. Main Result

In this section, we present the known values of $\gamma_p(Q_n)$, for $n \leq 7$.

Corollary 5 $\gamma_p(Q_n) = \lceil n/2 \rceil$, $n \in \{1, 2, 3, 4\}$ and $\gamma_p(Q_5) = 4$.

Proof. When $n \in \{1, 2, 3, 4, 5\}$, the lower bound is equal to the upper bound in Theorem 3. $\square$

The following 2 lemmas, we gave a PDS to derive the upper bounds of $\gamma(Q_n)$, respectively. For ease of reading, the following vertex label is represented as a decimal (i.e. vertex 110 is represented as vertex 6).

Lemma 6 $\gamma_p(Q_6) = 6$.

Proof. Let the PDS $S_p = \{0, 7, 8, 27, 45, 50\}$ for Q_6 .

By OR1, all vertices in $\{1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 15, 16, 18, 19, 23, 24, 25, 26, 31, 32, 34, 37, 39, 40, 41, 44, 47, 48, 51, 54, 58, 59, 61\}$ are observed. According the order of observed pair sequence ($\langle 3, 35 \rangle, \langle 5, 21 \rangle, \langle 11, 43 \rangle, \langle 13, 29 \rangle, \langle 15, 14 \rangle, \langle 10, 42 \rangle, \langle 12, 28 \rangle, \langle 18, 22 \rangle, \langle 6, 38 \rangle, \langle 19, 17 \rangle, \langle 1, 33 \rangle, \langle 16, 20 \rangle, \langle 4, 36 \rangle, \langle 17, 49 \rangle, \langle 20, 52 \rangle, \langle 21, 53 \rangle, \langle 22, 30 \rangle, \langle 14, 46 \rangle, \langle 23, 55 \rangle, \langle 24, 56 \rangle, \langle 25, 57 \rangle, \langle 28, 60 \rangle, \langle 30, 62 \rangle, \langle 31, 63 \rangle$), unobserved vertices will be observed by OR2.

The lower bound of $\gamma_p(Q_6)$ can be determined by Theorem 3. $\square$

Lemma 7 $\gamma_p(Q_7) = 10$.

Proof. Let the PDS $S_p = \{0, 7, 8, 9, 50, 53, 84, 95, 102, 121\}$ for Q_7 . By OR1, all vertices in $\{1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 15, 16, 18, 20, 21, 23, 24, 25, 31, 32, 34, 37, 38, 39, 40, 41, 48, 49, 51, 52, 54, 55, 57, 58, 61, 64, 68, 70, 71, 72, 73, 79, 80, 85, 86, 87, 89, 91, 92, 93, 94, 98, 100, 103, 105, 110, 113, 114, 116, 117, 118, 120, 123, 125, 127\}$ are observed. According the order of observed pair sequence ($\langle 2, 66 \rangle, \langle 4, 36 \rangle, \langle 5, 69 \rangle, \langle 16, 17 \rangle, \langle 21, 29 \rangle, \langle 25, 27 \rangle, \langle 29, 28 \rangle, \langle 20, 22 \rangle, \langle 6, 14 \rangle, \langle 15, 47 \rangle, \langle 22, 30 \rangle, \langle 23, 19 \rangle, \langle 17, 81 \rangle, \langle 19, 83 \rangle, \langle 28, 60 \rangle, \langle 31, 63 \rangle, \langle 36, 44 \rangle, \langle 12, 76 \rangle, \langle 38, 46 \rangle, \langle 14, 78 \rangle, \langle 39, 35 \rangle, \langle 3, 67 \rangle, \langle 34, 42 \rangle, \langle 46, 62 \rangle, \langle 30, 26 \rangle, \langle 10, 74 \rangle, \langle 18, 82 \rangle, \langle 26, 90 \rangle, \langle 27, 59 \rangle, \langle 49, 33 \rangle, \langle 1, 65 \rangle, \langle 32, 96 \rangle, \langle 33, 97 \rangle, \langle 51, 115 \rangle, \langle 55, 119 \rangle, \langle 57, 56 \rangle, \langle 24, 88 \rangle, \langle 40, 104 \rangle, \langle 48, 112 \rangle, \langle 58, 122 \rangle, \langle 59, 43 \rangle, \langle 11, 75 \rangle, \langle 35, 99 \rangle, \langle 41, 45 \rangle, \langle 13, 77 \rangle, \langle 37, 101 \rangle, \langle 42, 106 \rangle, \langle 43, 107 \rangle, \langle 44, 108 \rangle, \langle 45, 109 \rangle, \langle 47, 111 \rangle, \langle 60, 124 \rangle, \langle 62, 126 \rangle$), unobserved vertices will be observed by OR2.

The lower bound of $\gamma_p(Q_7)$ can be determined by Theorem 3. $\square$

As a concluding remark, we summarize Theorem 3, Corollary 5 and Lemmas 6, 7 as follows.

Theorem 8 The known values or bounds of $\gamma_p(Q_n)$ are shown as below.

- (1) $\gamma_p(Q_n) = \lceil n/2 \rceil, n \in \{1, 2, 3, 4\}$,
- (2) $\gamma_p(Q_5) = 4$,
- (3) $\gamma_p(Q_6) = 6$,
- (4) $\gamma_p(Q_7) = 10$,
- (5) $2^{n-1}/n \leq \gamma_p(Q_n) \leq 2^{n-\lfloor \log n \rfloor - 1}$, for $n \geq 8$, [6].

By Theorem 8, we disprove Conjecture 4.

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